

Worksheet builder

Lesson 01

1. Show that $f^{-1}(x) = (px + \sqrt{qx^2 + r})/4$, where p , q and r are integers. [4 marks]

2. Given that f^{-1} exists, write down the range of f^{-1} . [1 mark]

3. On the axes, sketch the graph of $y = |(x^2 - 4)(x - 2)|$, stating the intercepts with the coordinate axes. [4 marks]

4. Find the values of x for which y exists. [2 marks]

5. Using your answer to part (ii) state the relationship between the functions f and f^{-1} . [1 mark]

6. Given that $g(x) = (2x + 1)^{1/2} + 4$, for $x > 0$, solve the equation $gg(x) = 9$. [3 marks]

Answer key

1. $f^{-1}(x) = (3x + \sqrt{9x^2 + 8})/4$
2. $f^{-1} > 0$
3. Graph has maximum at $(-2/3, 256/27)$, $y=8$ intercept, $x=-2, 2$ intercepts, cusp at $x=2$
4. $x > -2$
5. f and f^{-1} are the same or f is self-inverse
6. $x = 31.5$